

Annotated slides

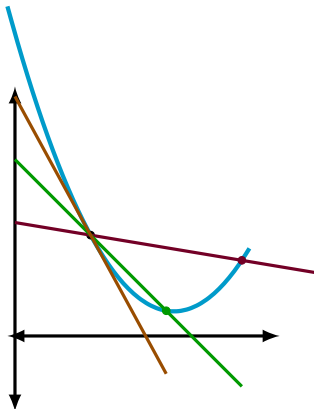
2727-MA140 Engineering Calculus

Week 04, Lecture 1 Introduction to Derivatives

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2026



What we'll study today

- 1 Remember:
 - Tutorials and assignments
 - Class Test
- 2 Derivatives: the concept
 - Rate of change
 - The definition
- Example
- 3 Derivative as a function
- 4 Differentiation by rule
 - 1. The Constant Rule
 - 2. The Power Rule
 - 3. The constant multiple rule
- 5 Exercises

Further reading:

- ▶ Sections 3.1 and 3.2 of **Calculus** by Strang & Herman:
<https://openstax.org/books/calculus-volume-1/pages/3-1-defining-the-derivative>
- ▶ Nice animation: <https://www.geogebra.org/m/MeMdCUEm>

Tutorials and Assignments

- ▶ **Assignment 1** finished yesterday at 5pm. Grades are available on Canvas
- ▶ **Problem Set 2** is open; deadline is 5pm, 5 Oct. To access it: go to Canvas, click on **Assignments ... Problem Set 2 ...** the bottom of the page, click **Load Problem Set 2 in a new window**. Or click on <https://universityofgalway.instructure.com/courses/58477/assignments/163696>
- ▶ This week's **Tutorial Sheet** is available at <https://universityofgalway.instructure.com/courses/58477/files/3838766>
- ▶ Beidh ranganna teagaisc MA140 nua (i nGaeilge), gach Dé Máirt ó 2 go 3, in AMB-G008 (Arts Millenium Building). Tosaíonn siad inniu.

Class Test

- ▶ Our first Class Test The first (of two) class tests will take place next Tuesday, 6th October.
- ▶ Last chance to request accommodations: fill out the form at <https://forms.cloud.microsoft/e/rBiq8RHHY2>
- ▶ Sample test to be posted tomorrow.

Derivatives: the concept

The next section of MA140 is all about **derivatives** of function. The derivative of a function describes how quickly the function is changing.

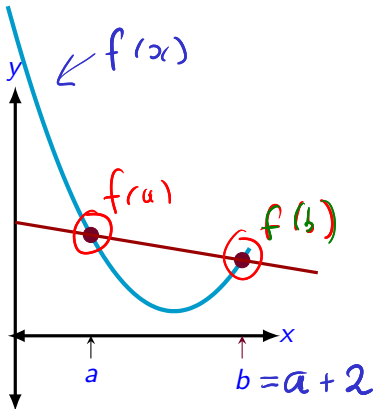
There are many, many applications: derivatives, and equations involving them are used everywhere: **speed/velocity** is the rate of change of displacement; **acceleration** is the rate of change of velocity.

We use derivatives to model how quickly a tumour is growing or shrinking, how pollutants are dispersed in a river, how pressure changes with depth, how inflation is changing in an economy. The list of applications is practically limitless.

Consider the graph opposite. It shows a function, f , and a secant line that intersects f at point $x = a$ and $x = b = a + 2$.

If we wanted to summarised how f is changing between those two values, we could compute it as

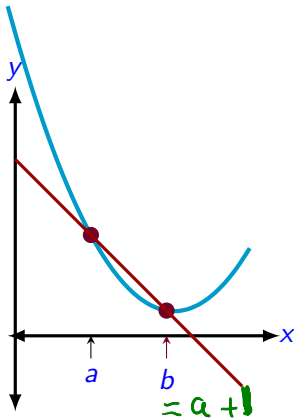
$$\frac{f(b) - f(a)}{b - a} = \frac{f(a + 2) - f(a)}{2}$$



Now we'll consider how f is changing over a shorter interval: from a to $b = a + 1$. Again, we sketch the secant line that intersects f at $x = a$ and $x = b$. The rate of change of f between these two values is

$$\frac{f(b) - f(a)}{b - a} = \frac{f(a + 1) - f(a)}{1},$$

which, of course, is the slope of the secant line.



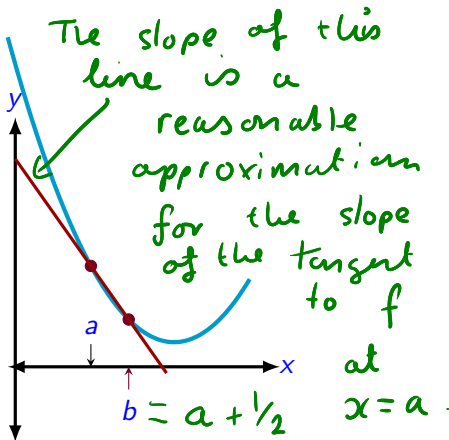
$$M = \frac{y_2 - y_1}{x_2 - x_1}$$

is the slope of the line joining (x_1, y_1) to (x_2, y_2)

Next we shorten interval again:
looking at how f changes from
 a to $b = a + \frac{1}{2}$, along with the
secant line that intersects f at
 $x = a$ and $x = b$.

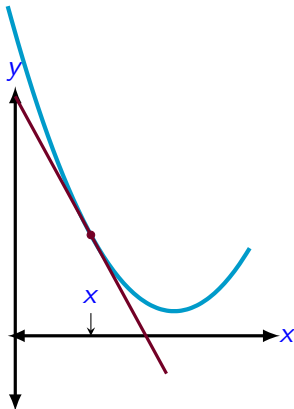
The rate of change of f between
these two values is

$$\frac{f(b) - f(a)}{b - a} = \frac{f(a + \frac{1}{2}) - f(a)}{\frac{1}{2}}.$$



Finally, suppose we want to looking at the **instantaneous** rate of change of f at $x = a$. Hopefully, the preceding images have convinced you we could do this in two (equivalent) ways:

1. $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$
2. or as the slope of the tangent to f at $x = a$.



That is, set $b = a + h$, then let $h \rightarrow 0$
 So $\frac{f(b) - f(a)}{b - a}$ is $\frac{f(a+h) - f(a)}{(a+h) - a}$

The slope of the curve $y = f(x)$ at the point $P = (a, f(a))$ is given by the number (if it exists)

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

This is called the **derivative of f at $x = a$** .

It is denoted as $f'(a)$ or $\frac{df}{dx}(a)$.

Definition: derivative at a point

Let $f(x)$ be a function that has $x = a$ in its domain. The **derivative** of the function $f(x)$ at a , denoted $f'(a)$, is defined by

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

if the limit exists.

We read " $f'(a)$ " as " f prime of a "
or "the derivative of f at a ", etc.

Some terminology

$f'(a)$ exists then we say that function f is differentiable at $x = a$.

Also, we'll soon learn that there are formulae for derivatives of many well-known functions. But when we use the “limit” formula, we are doing “**differentiation from first principles**”.

In some areas of mechanics / physics
if f is a function of time,
the derivative is written $\dot{f}$

Example

Use the limit definition of a derivative to compute the slope of the tangent to $f(x) = x^2$ at $x = 3$.

$$\begin{aligned} \text{We want to evaluate } & \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(3+h)^2 - 3^2}{h} = \lim_{h \rightarrow 0} \frac{9 + 6h + h^2 - 9}{h} \\ &= \lim_{h \rightarrow 0} \frac{(6+h)\cancel{h}}{\cancel{h}} = \lim_{h \rightarrow 0} 6+h = 6. \end{aligned}$$

Example

Use the limit definition of a derivative to find the equation of the tangent to $f(x) = 1/x$ at $x = 2$.

To find the tangent to f at $x = 2$, we need to know its slope: $f'(2)$

$$f'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \left(\frac{1}{h} \right) \left(\frac{1}{2+h} - \frac{1}{2} \right)$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{2}{(2+h)(2)} - \frac{2+h}{(2+h)h} \right) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{-h}{4+2h} \right)$$

$$= \lim_{h \rightarrow 0} \frac{-1}{4+2h} = -1/4. \quad \text{So } f'(a) = -1/4.$$

Example

Use the limit definition of a derivative to find the equation of the tangent to $f(x) = 1/x$ at $x = 2$.

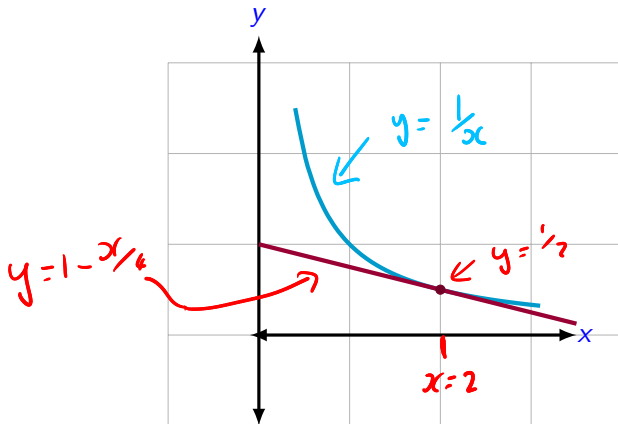
(Continued). So we now need the equation of the line with slope $-\frac{1}{4}$ through $(2, f(2)) = (2, \frac{1}{2})$.

Formula: $y - y_1 = m(x - x_1)$
 $\Rightarrow y - \frac{1}{2} = -\frac{1}{4}(2 - x)$

which is

$$y = 1 - \frac{x}{4}$$

$$f(x) = \frac{1}{x} \quad \text{and} \quad y = 1 - \frac{x}{4}$$



Derivative as a function

We've seen how to compute $f'(a)$: the derivative of the function f at a given point, $x = a$.

But if $f'(a)$ has a value for all $x = a$ (in the domain of $f(x)$), we can think $f'(x)$ as a function itself!

Definition: derivative as a function

Let f be a function. The derivative function, denoted f' , is the function whose domain consists of those values of x such that the following limit exists:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Derivative as a function

Terminology and notation

- ▶ We usually refer to f' simply as the derivative of $f(x)$.
- ▶ Where $y = f(x)$, we often write f' as $\frac{dy}{dx}$, or y' , or $\frac{d}{dx}(f)(x)$.

Note f' is same as $f'(x)$.

$$\frac{df}{dx} \quad \text{or} \quad \frac{df}{dx}(x)$$

Derivative as a function

Example

Use the above definition to find the derivative of $f(x) = x^2$

Solution

The derivative is defined as follows:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Here $f(x+h) = (x+h)^2 = \underline{x^2} + \underline{h^2} + \underline{2hx}$, so we get:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\cancel{x^2} + h^2 + 2hx - \cancel{x^2}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{h}(h+2x)}{\cancel{h}} = \lim_{h \rightarrow 0} (h+2x) = 2x \end{aligned}$$

Derivative as a function

Example

Use the "limit" definition to show that the derivative of $f(x) = \sqrt{x}$ is $f'(x) = \frac{1}{2\sqrt{x}}$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

we "complete the square" $\frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})}$

$$= \frac{(\sqrt{x+h} \sqrt{x+h}) + \cancel{\sqrt{x+h} \sqrt{x}} - \cancel{\sqrt{x} \sqrt{x+h}} - (\sqrt{x} \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})}$$

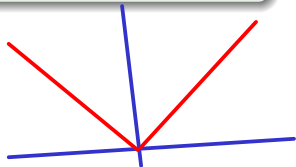
$$= \frac{\cancel{h}(\cancel{x+h} - x)}{h(\sqrt{x+h} + \sqrt{x})} = \frac{\cancel{h}}{h(\sqrt{x+h} + \sqrt{x})}$$

$$\text{So } f'(x) = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}} \quad \checkmark$$

Derivative as a function

Consider the absolute value function $f(x) = |x|$. What is its derivative at (i) $x = 2$, (ii) $x = -3$, or (iii) $x = 0$?

Recall' $f(x) = |x| = \begin{cases} -x & x < 0 \\ x & x \geq 0 \end{cases}$



$$(i) \quad f'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{2+h-2}{h} = 1 \quad \checkmark$$

$$(ii) \quad f'(-3) = -1 \quad (\text{check!})$$

$$(iii) \quad f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}, \text{ which does not exist, since } \lim_{x \rightarrow 0^-} f(x) = -1 \neq 1 = \lim_{x \rightarrow 0^+} f(x)$$

Derivative as a function

Show that $\frac{d}{dx}(\sin x) = \cos x$.

Solution: We need to evaluate

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h},$$

where $f(x) = \sin(x)$. From p5 of the “log” tables, we have that

$$\sin(A) - \sin(B) = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right).$$

Here $A = x + h$, and $B = x$, so

$$\sin(x+h) - \sin(x) = 2 \cos\left(\frac{2x+h}{2}\right) \sin\left(\frac{h}{2}\right).$$

So now we evaluate

$$f'(x) = \lim_{h \rightarrow 0} \frac{2 \cos\left(\frac{2x+h}{2}\right) \sin\left(\frac{h}{2}\right)}{h} = \lim_{h \rightarrow 0} \frac{2}{h} \sin\left(\frac{h}{2}\right) \cos\left(\frac{2x+h}{2}\right).$$

Derivative as a function

But

$$\lim_{h \rightarrow 0} \frac{2}{h} \sin\left(\frac{h}{2}\right) \cos\left(\frac{2x+h}{2}\right) = \left(\lim_{h \rightarrow 0} \frac{2}{h} \sin\left(\frac{h}{2}\right) \right) \left(\lim_{h \rightarrow 0} \cos\left(\frac{2x+h}{2}\right) \right).$$

We learned last week that,

$$\lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta} = 1.$$

Finished here!

Taking $\theta = h/2$, we get that

$$\lim_{h \rightarrow 0} \frac{2}{h} \sin\left(\frac{h}{2}\right) = 1.$$

And finally,

$$\lim_{h \rightarrow 0} \cos\left(\frac{2x+h}{2}\right) = \cos(x).$$

and we are done!

Differentiation by rule

We've seen we can compute derivatives of some functions using the “limit” definition (i.e., **differentiation from first principles**). However, that approach is tedious, and unnecessary in many case.

Instead we can use a set of “**rules**” which makes the process much more efficient. These rules are themselves derived from the “limit” definition – but we don't have to use that every time.

The Constant Rule

If f is a constant function, i.e. $f(x) = c$ for all x , then:

$$\frac{df}{dx} = \frac{d}{dx}(c) = 0.$$

Why:

We've already deduced that

- ▶ The derivative of $f(x) = x^2$ is $f'(x) = 2x$
- ▶ The derivative of $f(x) = x^{1/2}$ is $f'(x) = \frac{1}{2}x^{-1/2}$

These are particular examples of the **Power Rules**

The Power Rule

If n is any real number, then

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

Examples Calculate the derivatives of the following functions

1. $f(x) = x^6$

2. $f(x) = \sqrt[3]{x}$

The constant multiple rule

Let $f(x)$ be any differentiable function, and let k be constant, then

$$\frac{d}{dx}(kf(x)) = k \frac{d}{dx}(f(x)).$$

Example: Find the derivative of $f(x) = 5x^4$.

Exercises 4.1.1 (Based on Q2(a), 2019/2020)

Use the (limit) definition of a derivative to differentiate the function $f(x) = x^2 + 2$.

Exercise 4.1.2

Use the (limit) definition of a derivative to show that the derivative of $f(x) = \cos(x)$ is $f'(x) = -\sin(x)$.